

Motor Datasheet

Operating Point: Continuous

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	280.0	mm
Machine Inner Diameter	160.0	mm
Machine Thickness	88.0	mm
Airgap	1.0	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	18.55	kg
Rotor Mass	2.10	kg
Stator Mass	16.45	kg
Rotor Moment of Inertia (active elements only) - incl. simplified rotor structure	0.0449	kg m2

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	299.8	A
Phase Resistance	9.2	mOhm
D-Axis Inductance (Ld)	165.6	μH
Q-Axis Inductance (Lq)	156.5	μH
PM Flux Linkage	88.8	mWb
Frequency at operating point (electrical)	200.0	Hz
Winding Configuration	wye (star)	-

Performance

Parameter	Value	Unit
Torque (Continuous)	319.77	Nm
Speed	2000.0	rpm
BackEMF Constant (Ke)	58.6	mV/rpm
Motor Constant (Km, only resistive losses)	6.13	Nm/sqrt(W)
Torque Constant (Kt, no-load)	799.2	mNm/A_peak
Saturation	0.0	%

Losses

Parameter	Value	Unit
Total Losses	2824.3	W
Resistive Losses	2469.5	W

Configuration

Parameter	Value	Unit
Number of Stators	2	-
Number of Rotors	1	-
Number of Slots	18	-
Pole Pairs	6	-
Number of Parallel Paths	4	-
Number of Turns per Tooth (designed incl. parallel paths)	24	-
Number of Phase Turns (all phases in series)	72	-